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Heuristic Search in Machine Learning

ITA0612 : Machine Learning

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What is heuristic search?

- A search strategy that uses an estimate of how promising each option is
- Explores the most promising candidates first, instead of blindly
- Trades guaranteed optimality for speed, unless the heuristic is admissible

Uninformed: explores everything

Heuristic: explores what looks best

$$f(n) = g(n) + h(n)$$

$g(n)$ cost paid so far

$h(n)$ estimated cost to the goal

$f(n)$ estimated total, used to rank nodes



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Why ML needs heuristics

The hypothesis space explodes. Boolean functions over n binary features: $2^{(2^n)}$

$n = 3$

256

possible hypotheses

$n = 5$

4.3 billion

possible hypotheses

$n = 6$

1.8×10^{19}

possible hypotheses

Enumerating every hypothesis is infeasible, so learning needs a guide.



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Exploring the hypothesis space

Most general

$\langle ?, ?, ? \rangle$



Partially specific

$\langle \text{Sunny}, ?, ? \rangle$



Most specific

$\langle \text{Sunny}, \text{Warm}, \text{Strong} \rangle$

- Hypotheses are ordered from general to specific
- Search moves along this ordering, specializing or generalizing
- Training examples prune whole regions at once
- The version space holds every hypothesis still consistent with the data



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How heuristics guide the search

1

Ordering

Expand the most promising candidate first

2

Pruning

Discard branches that cannot beat the best so far

3

Evaluation function

Score candidates: accuracy, information gain, loss

4

Inductive bias

Prefer simpler or more general hypotheses (Occam)



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Common heuristic search strategies



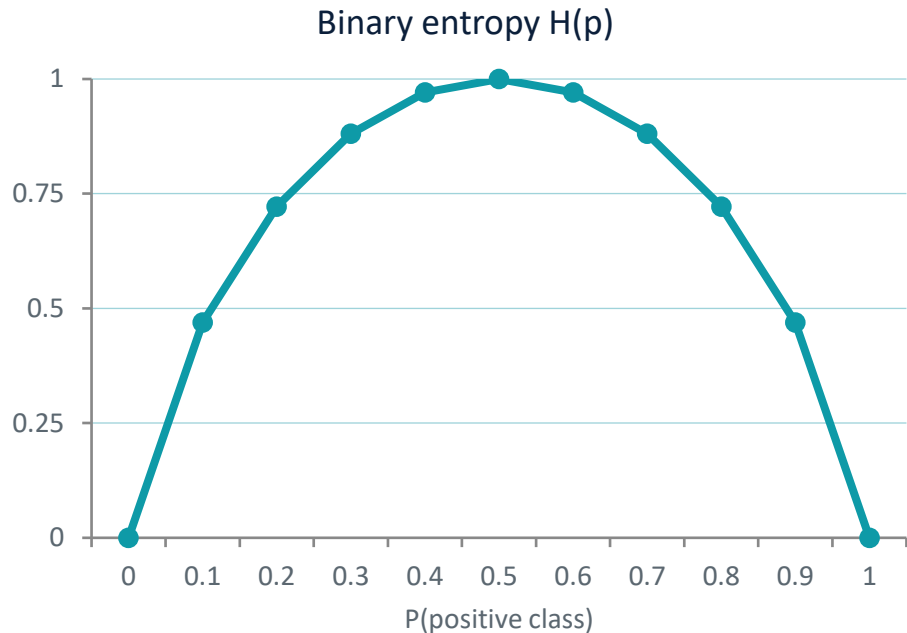
Strategy	Heuristic idea	ML use
Greedy / hill climbing	Take the locally best step	Decision tree splits
Best-first / A*	Rank by $f(n) = g(n) + h(n)$	Structure learning
Beam search	Keep only the top-k candidates	Sequence decoding
Simulated annealing	Accept worse moves, less over time	Hyperparameter tuning
Genetic algorithms	Evolve a population by fitness	Feature / architecture search



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Example 1: decision trees



- At each node, greedily pick the split with the highest information gain
- Pure nodes (entropy 0) need no further search
- Builds a good tree in polynomial time instead of searching all trees



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Example 2: feature selection

Exhaustive search

1,048,576

feature subsets to evaluate (2^{20})

Greedy forward selection

210

models to evaluate ($20 \cdot 21 / 2$)

About 5,000× fewer evaluations. Add the feature that helps most, repeat. Backward elimination and RFE follow the same idea.



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Example 3: more applications

1

Hyperparameter tuning

Random search, successive halving and evolutionary methods spend compute on promising configurations

2

Architecture search

Evolutionary or learned controllers propose networks, and a cheap proxy score guides the next move

3

Beam search decoding

Keep the top-k partial sequences at each step instead of all possible outputs



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Performance gains and trade-offs



+

What we gain

- Exponential cost reduced to polynomial
- Faster training and model selection
- Good solutions on huge spaces

!

What we give up

- No guarantee of the global optimum
- Local optima and plateaus
- Results depend on heuristic quality



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Conclusion



- **Heuristics make the search practical.** By ordering candidates, pruning unpromising branches, and scoring options with an evaluation function, they steer the search toward promising regions.
- **Inductive bias narrows the space.** Preferring simpler or more general hypotheses (Occam's razor) limits what the learner has to consider.
- **Real ML algorithms already rely on heuristics.** Greedy splits in decision trees, forward selection for features, beam search in decoding, and evolutionary or annealing-style tuning all use heuristic guidance.
- **The gain is large.** Exponential cost often drops to polynomial. In the feature selection example, 210 evaluations replace over a million.

*Thank
you!*